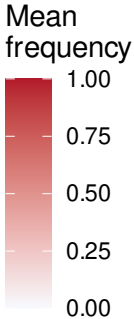


Consensus in silico perturbation screen

Mean across 5 computational seeds; intervals are provided in Table 02

|                                            |      |      |      |      |      |
|--------------------------------------------|------|------|------|------|------|
| BAX E1 + GSDME KO                          | 0.00 | 0.07 | 0.91 | 0.00 | 0.01 |
| BAX E1                                     | 0.07 | 0.00 | 0.93 | 0.00 | 0.01 |
| BCL2 KO                                    | 0.06 | 0.00 | 0.90 | 0.00 | 0.04 |
| CASP3 KO                                   | 0.00 | 0.00 | 0.90 | 0.00 | 0.05 |
| GSDME KO + CASP3 E1                        | 0.00 | 1.00 | 0.92 | 0.00 | 0.00 |
| CASP3 E1                                   | 1.00 | 0.00 | 0.91 | 0.00 | 0.00 |
| GSDME E1 + CASP3 E1                        | 1.00 | 0.00 | 0.91 | 0.00 | 0.00 |
| GSDME KO                                   | 0.00 | 0.04 | 0.93 | 0.00 | 0.04 |
| GSDME E1                                   | 1.00 | 0.00 | 0.91 | 0.00 | 0.04 |
| MALAT1 E1 + miR-204-5p KO + SIRT1 E1       | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 |
| MALAT1 KO + GSDME KO                       | 0.00 | 0.07 | 0.85 | 0.00 | 0.07 |
| MALAT1 KO                                  | 0.09 | 0.00 | 0.85 | 0.00 | 0.06 |
| MALAT1 E1                                  | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 |
| miR-204-5p KO                              | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 |
| miR-204-5p E1 + CASP3 KO                   | 0.00 | 0.00 | 0.00 | 0.00 | 0.48 |
| miR-204-5p E1 + GSDME KO                   | 0.00 | 0.49 | 0.00 | 0.00 | 0.51 |
| miR-204-5p E1                              | 0.47 | 0.00 | 0.00 | 0.00 | 0.53 |
| p21 E1                                     | 0.01 | 0.00 | 0.89 | 0.00 | 0.10 |
| p53 E1 + miR-204-5p E1 + CASP3 KO + p21 E1 | 0.00 | 0.00 | 0.00 | 0.00 | 1.00 |
| p53 E1                                     | 0.25 | 0.00 | 0.59 | 0.00 | 0.16 |
| p53 KO                                     | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 |
| Reference: DDR ON, GSDME endogenous        | 0.05 | 0.00 | 0.91 | 0.00 | 0.04 |
| SIRT1 KO + GSDME KO + CASP3 E1             | 0.00 | 1.00 | 0.00 | 0.00 | 0.00 |
| SIRT1 KO + GSDME KO                        | 0.00 | 0.92 | 0.00 | 0.00 | 0.08 |
| SIRT1 KO                                   | 0.92 | 0.00 | 0.00 | 0.00 | 0.08 |
| SIRT1 E1                                   | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 |



GSDME-mediated pyroptosis

Apoptosis

Resistance

Proliferation

Cell-cycle arrest